Introduction

Bayes' theorem is the rule for inverting conditional probabilities: given $P(B \mid A)$, it tells us $P(A \mid B)$. It is the mathematical bridge between diagnostic sensitivity and positive predictive value, between forward simulation and inverse inference, between a likelihood function and a posterior distribution. Every Bayesian analysis, every clinical prediction, and every probabilistic classifier is an application of Bayes' theorem in disguise.

Prerequisites

Conditional probability and the law of total probability.

Theory

For events A and B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}.$$

When $\{A_i\}$ forms a partition of the sample space, the denominator expands by the law of total probability:

$$P(A_i \mid B) = \frac{P(B \mid A_i)\, P(A_i)}{\sum_j P(B \mid A_j)\, P(A_j)}.$$

In the Bayesian vocabulary:

- $P(A)$ is the **prior**: what was believed before observing B.
- $P(B \mid A)$ is the **likelihood**: how likely B is under each hypothesis.
- $P(A \mid B)$ is the **posterior**: updated belief about A after observing B.
- $P(B)$ is the **marginal likelihood** or "evidence".

Posterior $\propto$ likelihood $\times$ prior is the single most important slogan in Bayesian statistics.

Assumptions

- $P(B) > 0$; otherwise conditioning is undefined.
- The prior $P(A)$ must be specified; choosing it is where domain knowledge enters.

R Implementation

```r
library(ggplot2)

# Diagnostic test: prior = prevalence, likelihood = sensitivity and specificity
sensitivity <- 0.92
specificity <- 0.97
prevalences <- seq(0.001, 0.5, length.out = 100)

ppv <- sensitivity * prevalences /
       (sensitivity * prevalences + (1 - specificity) * (1 - prevalences))

df <- data.frame(prevalence = prevalences, ppv = ppv)

ggplot(df, aes(prevalence, ppv)) +
  geom_line(colour = "#2A9D8F", linewidth = 1) +
  geom_vline(xintercept = 0.01, linetype = "dashed", colour = "#F4A261") +
```

```r
  geom_vline(xintercept = 0.30, linetype = "dashed", colour = "#F4A261") +
  labs(x = "Disease prevalence",
       y = "Positive predictive value",
       title = "PPV depends strongly on prevalence (Bayes' theorem)") +
  theme_minimal()

c(prevalence_001 = ppv[which.min(abs(prevalences - 0.01))],
  prevalence_030 = ppv[which.min(abs(prevalences - 0.30))])
```

Output & Results

```
prevalence_001 prevalence_030
        0.2368         0.9290
```

At 1% prevalence, a positive test on a 92%-sensitive, 97%-specific assay gives only 24% probability of disease. At 30% prevalence, the same test yields a 93% PPV. The test characteristics are unchanged; the posterior depends crucially on the prior.

Interpretation

The prevalence dependence of PPV is the canonical real-world application of Bayes' theorem. Reporting "our test has 92% sensitivity and 97% specificity" is meaningless without the prevalence of the intended use population; the clinically relevant quantity is PPV.

Practical Tips

- Never cite sensitivity / specificity in isolation when discussing diagnostic utility; always give the PPV and NPV at the target prevalence.
- Bayes' theorem scales trivially to many hypotheses via the partition form; for continuous parameter spaces, the sums become integrals.
- Prior choice is contentious in some Bayesian applications; weakly informative priors (Cauchy, normal with wide scale) are standard defaults.
- Updating sequentially: the posterior after observation 1 becomes the prior for observation 2. Chains of updates are equivalent to a single update with all observations.
- The **likelihood ratio** $P(B \mid A)/P(B \mid A^c)$ converts the prior odds of A to the posterior odds multiplicatively – often a cleaner formulation than raw probabilities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots